



Become a Solution Architect

Enhance your Technical and Behavioural Skills

WHAT DID YOU
LEARN SO FAR
THIS YEAR?

Agenda

Topic	Duration	Presenter
Program Introduction	10 Mins	Prasad & Ashish
Team Introduction	5 Mins	All
Technical Topic	40 Mins	Ashish & Parna
Keynote	30 Mins	Prasad & Matt Wood
Workshop Walkthrough	30 Mins	Parna

Pre Launch Q&A Session

BeSA Pre-Launch

«« How to get most out of BeSA sessions? »»

-  **Schedule & Timings**
 - Plan ahead. Mark your calendar. Be on time.
-  **Badges & Certificates**
 - Earn, showcase and grow your cloud skills.
-  **How to Register for AWS Workshops (Hands-on)**
 - Step-by-step guide to secure your spot.

19 Aug
1 PM GMT

 **JOIN THE LIVE SESSION >>**



Become a Solutions Architect



Topic covered in previous batches

Batch	Topics - Technical Track	Topics – Behavioural Track
Batch 01	Cloud Computing Foundation	Becoming a Solutions Architect
Batch 02	Serverless	Crucial soft skills for a Solutions Architect
Batch 03	Networking in Cloud	Career Journeys
Batch 04	Fundamental IT Concepts	A day in the life of (TAM/SA/SDE/AM etc.)
Batch 05	Certification Track Practitioner and SA Asso.	Interview Preparation
Batch 06	Generative AI on AWS Fundamentals	Common interview questions
Batch 07	Gen AI – Hands on workshop	End-to-end hiring process of a SA at AWS
Batch 08	Migration and Modernization on AWS	Essentials skills for next generation SAs
Batch 09	Agentic AI on AWS	Developing SA mindset in AI era

Detailed Agenda

Date	Technical Track Agentic AI: POC to Production on AWS	Behavioural Track Leading Customer Discussions as an SA
Week 1 22-Aug	Anatomy of an AI Agent	Fireside chat with Matt Wood, Chief AI & Technology Officer, AWS
Week 2 29-Aug	Building Agentic AI with AWS Serverless	Why Facilitation is the SA Superpower
Week 3 5-Sep	Building AI Agents on Amazon EKS	Designing Workshops that Drive Outcomes
Week 4 12-Sep	Design Authentication, Authorization, and Logging	Facilitating Technical Decisions in the Room
Week 5 19-Sep	Agentic AI Evaluation and Observability	Visual Facilitation & Real-Time Architecture
Week 6 26-Sep	AI Governance on AWS	From Workshop to Action

Session Details

Technical Track



60

Mins

Agentic AI:
POC to
Production
on AWS

Behavioural Track



30

Mins

Leading
Customer
Discussions
as an SA

Workshop Walkthrough



30

Mins

Agentic AI
Workshops

Active Duration of Workshop

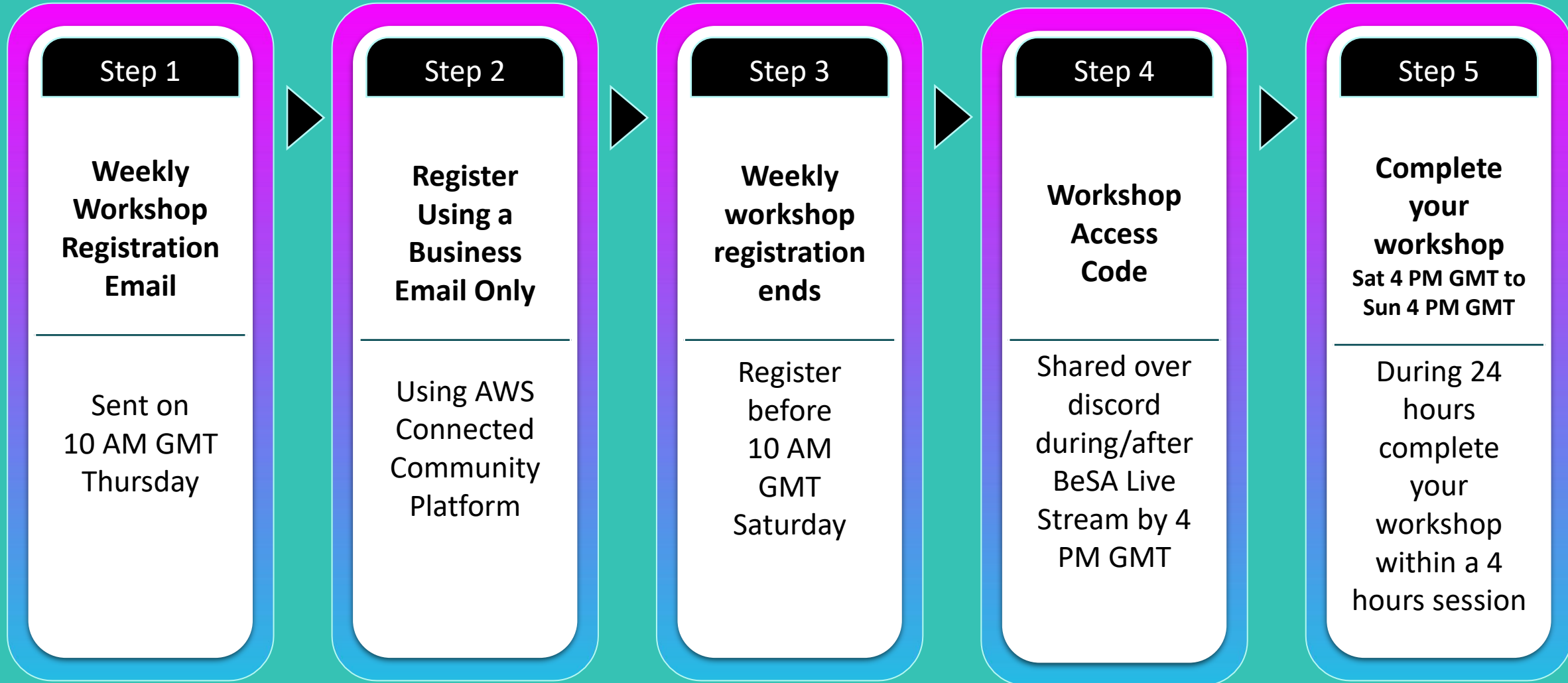


24

Hours

During 24 hours
complete your
workshop within
a 4 hours session

Workshop Flow



Earn badges upon completing each weekly Knowledge Check

Week 1



Week 2



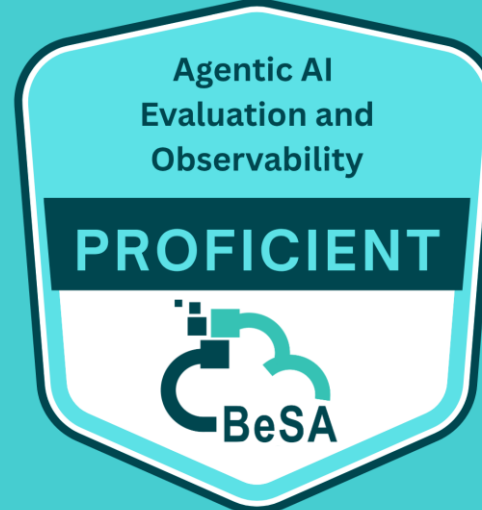
Week 3



Week 4



Week 5



Week 6



Earn a certificate upon passing all 6 Knowledge Checks



Meet the team





Anatomy of an
AI Agent

Agent

- Agent word comes from Latin word “*agere*”, which means
 - to do, to drive, to lead, to act
- Agent (as we know in real-world)
 - Someone who acts on behalf of another person or business.
- Typical Agents in real-life
 - Customer Support Agent
 - Insurance Agent
 - Real Estate Agent
 - Travel Agent



Analogy



Agent 007
Bond Franchisee



Agent Smith
The Matrix



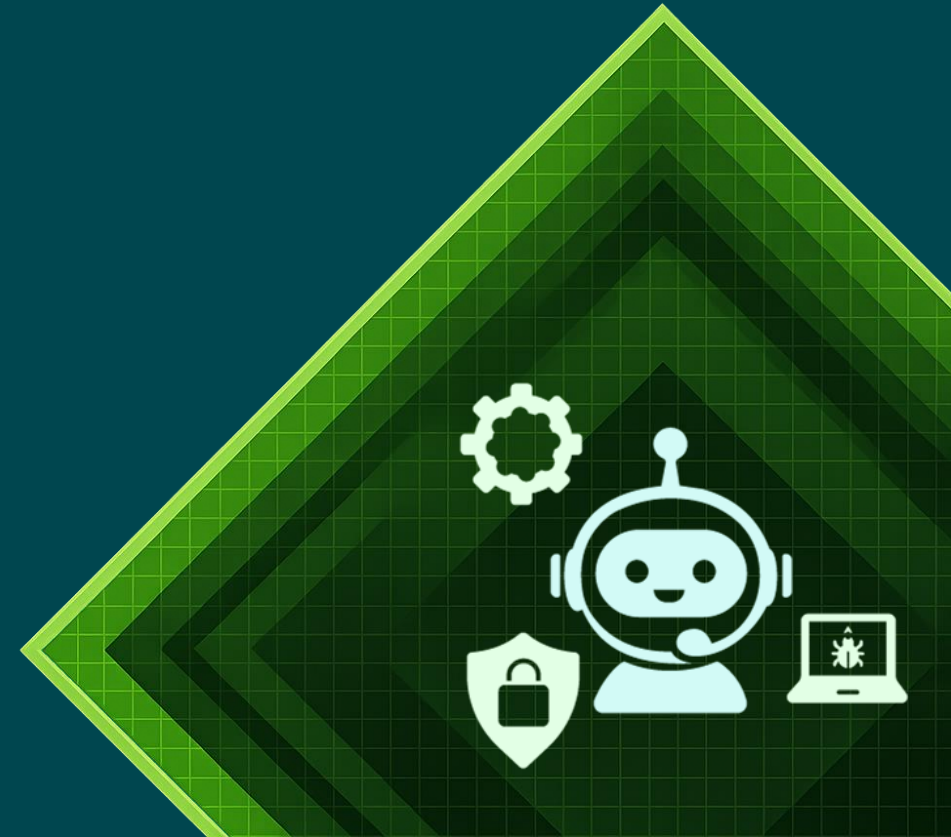
Agent 47
Hitman



Agent Dominika
Red Sparrow

AI Agent

- A software system that acts on behalf of a user or organization by using AI to make decisions and perform tasks.
- An AI Model acts as the reasoning engine of an AI Agent, but an AI Model alone is not an AI Agent.
- Typical AI Agents
 - Customer Support AI Agent
 - Travel Booking AI Agent
 - Research AI Agent
 - Financial Analysis AI Agent



Analogy



Agent 007



AI Agent

Receives a high-stakes mission from MI6

Assesses the threat

Plans covert action in real-time

Acts autonomously under pressure

Adapts to changing threats mid-operation

Saves the day with precision and skill

Triggered by a defined goal or user request

Analyzes context, data, and environment

Generates a plan using reasoning and tool access

Executes tasks across services without guidance

Replans based on new inputs or system feedback

Delivers outcomes with speed, accuracy, and scale

Agentic AI

- A complete AI system that enables AI Agents to autonomously achieve goals through planning, reasoning, memory, tool usage, and adaptive decision-making.
- Uses one or more AI Agents, powered by AI Models, to autonomously achieve goals.
- Typical Agentic AI Systems
 - Autonomous Customer Service Platform
 - Enterprise Research Assistant
 - IT Operations Automation Platform
 - AI Travel Planning Platform



AI Agent vs. Agentic AI

AI Agent

A single autonomous software entity.

Analogy:

AI Agent = Employee

- Has a role
- Performs tasks
- Makes decisions within its responsibility



IRONMAN

Agentic AI

A complete system or architecture built around autonomous agents.

Analogy:

Agentic AI = Organization

- Multiple employees
- Processes
- Knowledge
- Tools
- Coordination
- Shared goals



AVENGERS

AI Agent vs. Agentic AI

AI Agent	Agentic AI
A single autonomous software entity.	A complete system or architecture built around autonomous agents.
AI Agent = AI Model + Goal + Tools + Actions	Agentic AI = AI Model(s) + Agent(s) + Planning + Memory + Tool Calling + Orchestration + Guardrails
Performs tasks to achieve a specific goal.	Coordinates agents, memory, tools, workflows, and reasoning to achieve business outcomes.
Typically powered by an AI Model.	Typically contains one or more AI Model-powered agents.
Focuses on execution of a task.	Focuses on end-to-end problem solving.
Building block.	Complete solution.
Example: Customer Support Agent.	Example: Customer Support Platform with multiple agents, memory, RAG, guardrails, and orchestration.

When NOT to Use an Agent

When NOT To Use An Agenet

- The question is not "*Can an agent do this?*" - it's "*Should an agent do this?*"
- Three specific situations where a workflow or simpler system is the better choice:

The task is fully definable by rules.



Expense approval: If expense $\leq$ £500 → approve; otherwise → manager approval.

Latency is the primary constraint



Emergency stop: A machine detects danger → stop immediately

Full explainability is required out of the box.

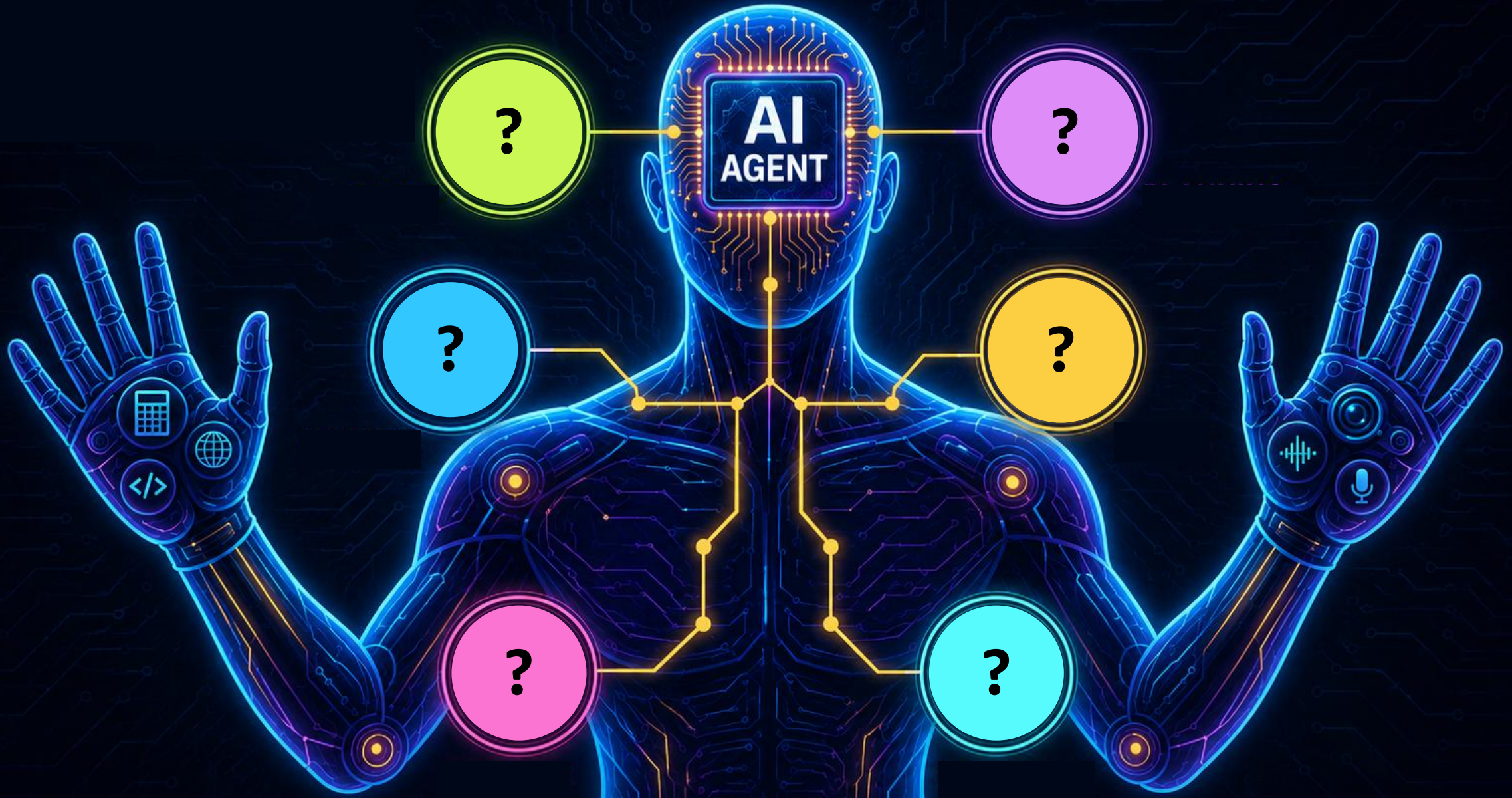


Loan eligibility: Approve/reject based on predefined income and credit-score rules.



Anatomy Of An AI Agent

Anatomy Of An Agent



Anatomy Of An Agent

MEMORY

What do I already know?

- Maintains context
- Stores prior interactions
- Retains knowledge and preferences

Types commonly referenced in Agentic AI:
Short-Term Memory
Long-Term Memory

GOALS

What should I achieve?

- Defines the desired outcome
- Provides direction for decision making
- Drives planning and action selection

Example: Resolve a customer complaint

PLANNING

How should I achieve it?

- Breaks goals into tasks
- Determines next best action
- Adapts decisions based on context

Example: Investigate order status before issuing a refund

TOOLS

What capabilities can I use?

- APIs
- Databases
- Search systems
- Calculators
- Enterprise applications

Example: Query CRM, or order systems

ACTIONS

What should I do next?

- Execute tasks
- Invoke tools
- Interact with users and systems

Examples:

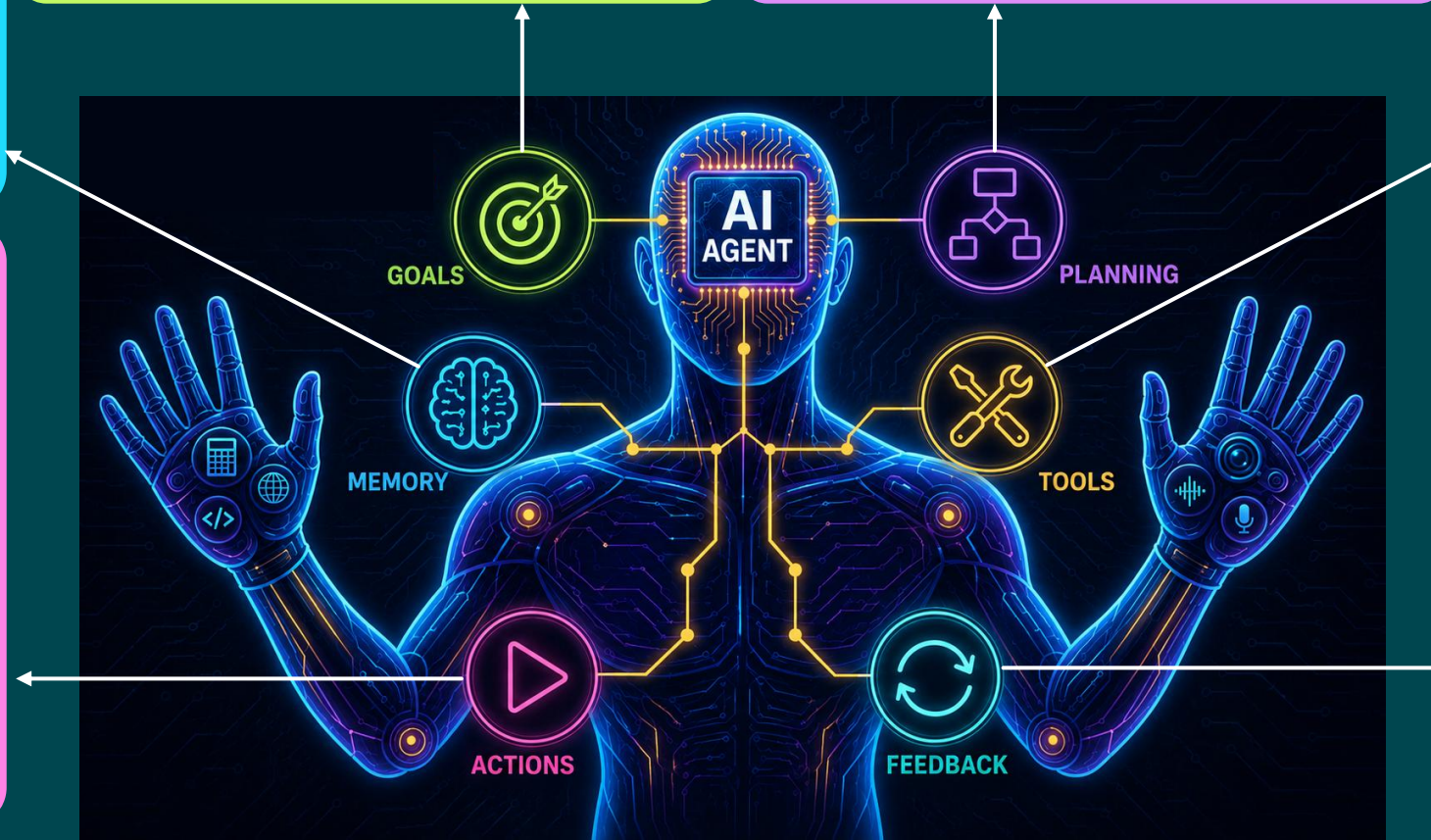
Send email
Create ticket
Update CRM
Query database

FEEDBACK

Did it work?

- Evaluates outcomes
- Observes environment changes
- Supports self-correction and replanning

Example: If inventory lookup fails, try another source

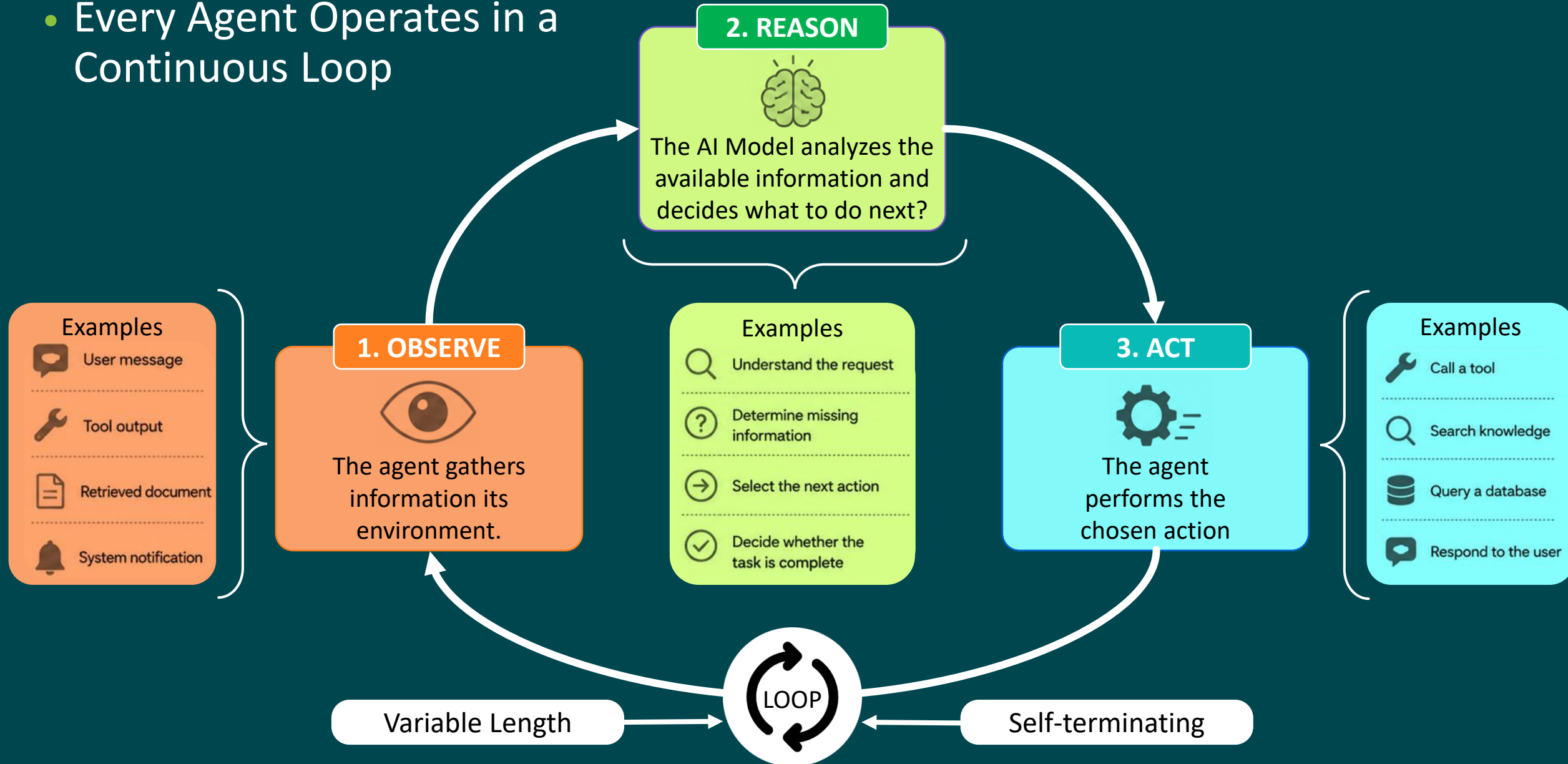




The Agent Loop

The Agent Loop - How an Agent Works

- Every Agent Operates in a Continuous Loop

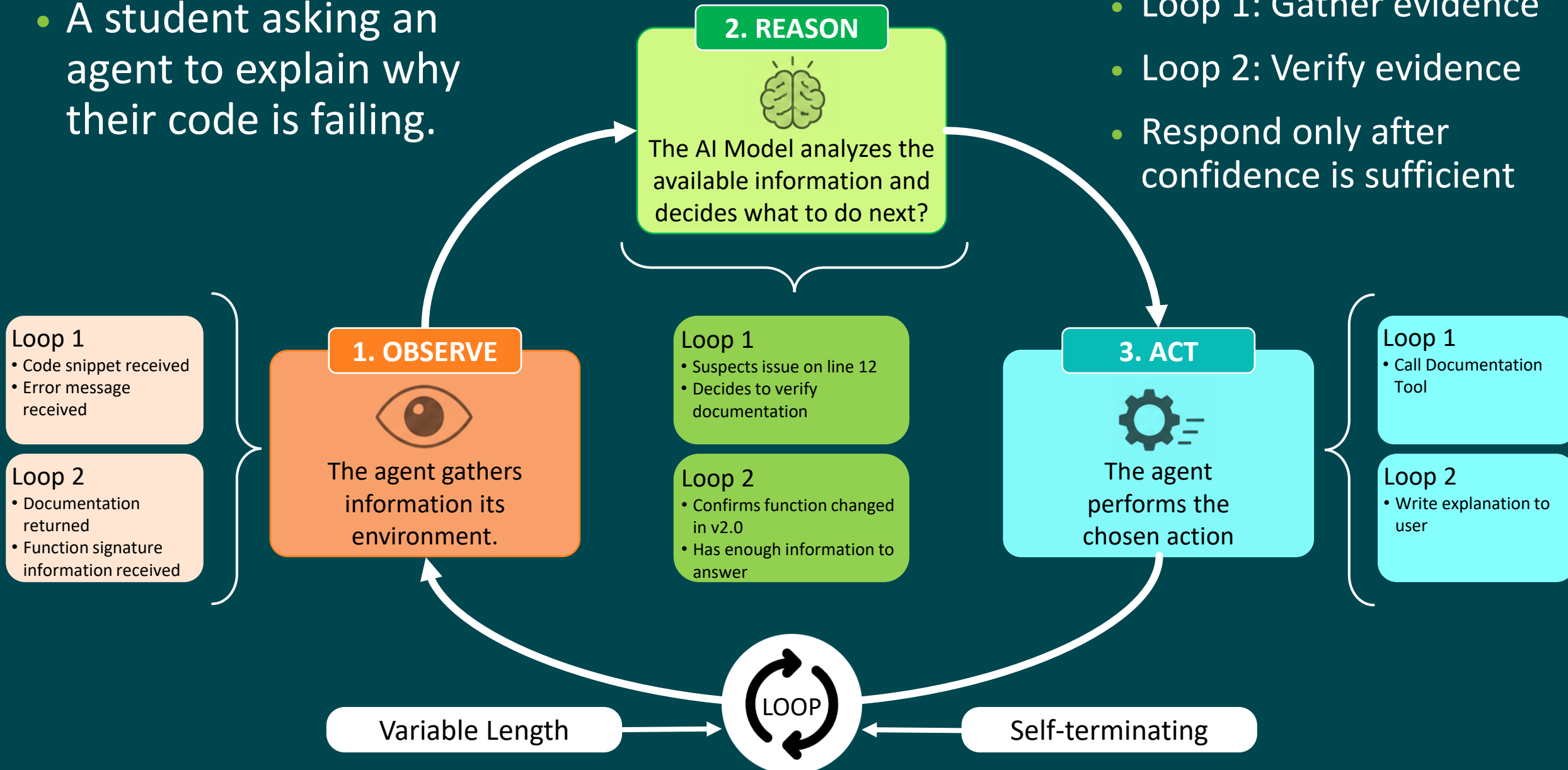


The Agent Loop - How an Agent Works

- A student asking an agent to explain why their code is failing.

Agent behaviour

- Loop 1: Gather evidence
- Loop 2: Verify evidence
- Respond only after confidence is sufficient



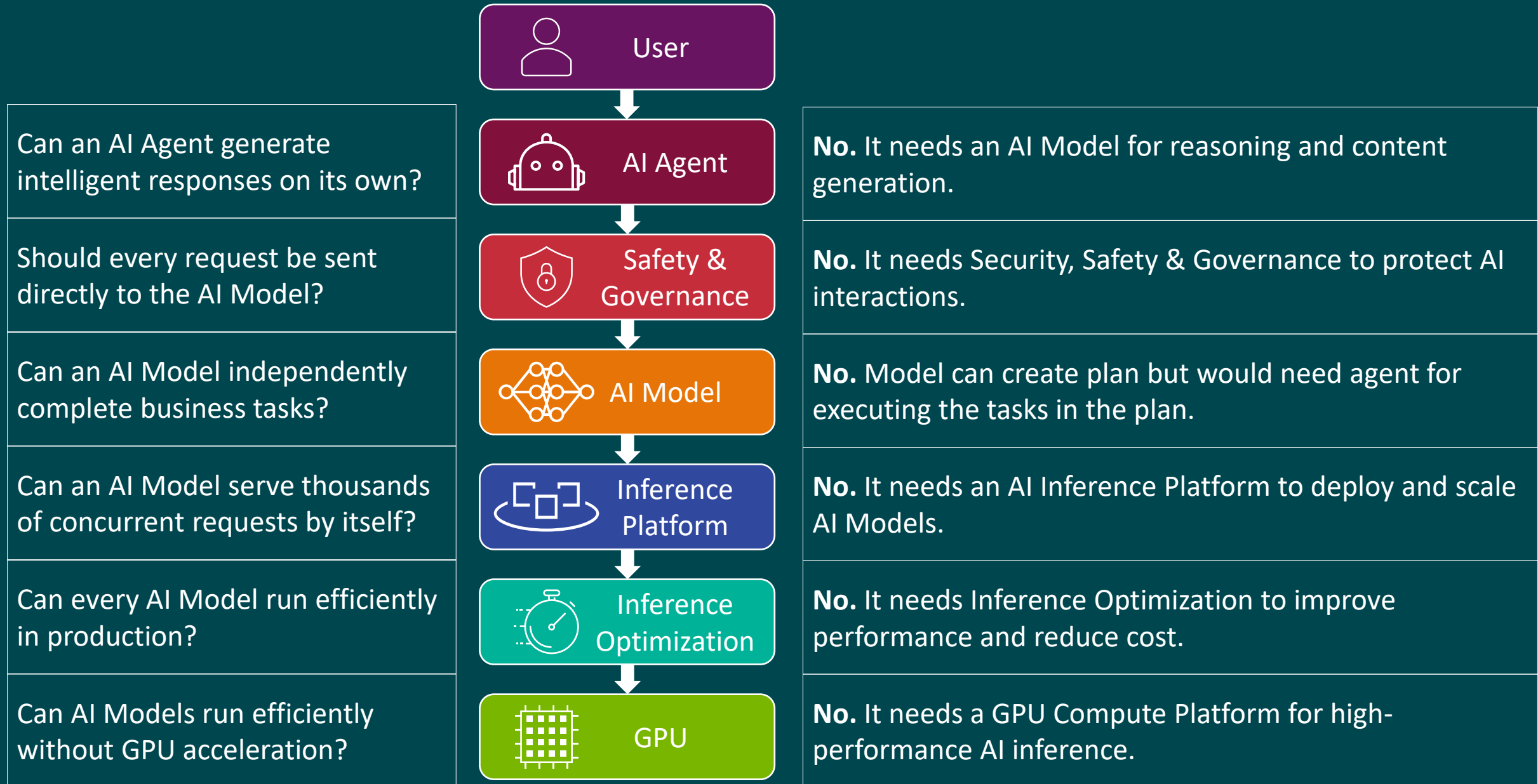


What Does an AI
Agent Need
Before It Can Run
in Production?

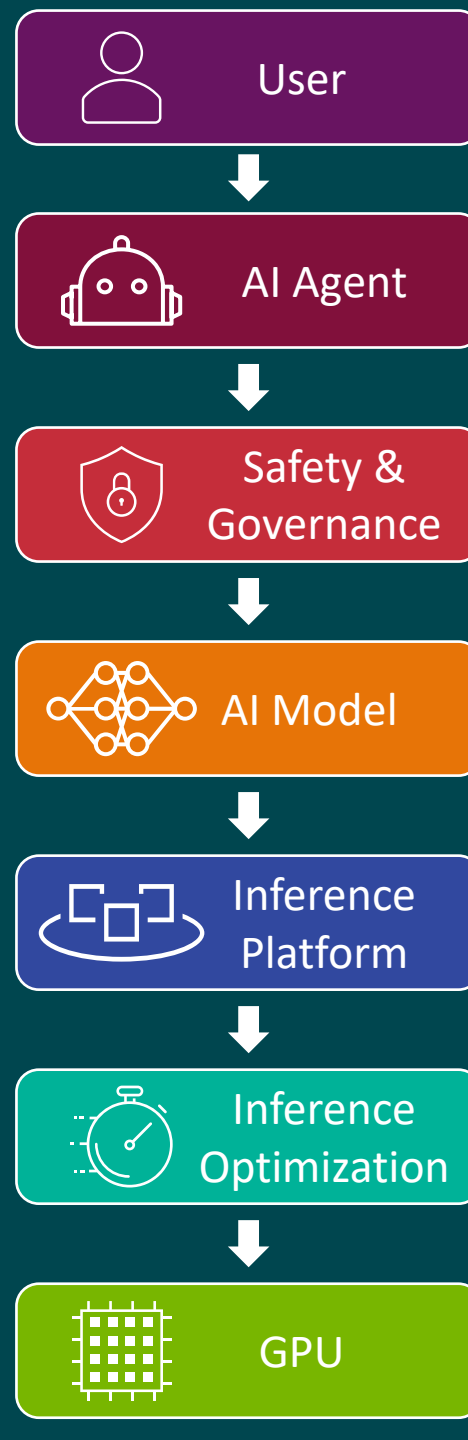
POC to Production: Top 5 Challenges

Challenge	POC	Production challenge
1. Scale & Performance	Works for a few test users	Must handle many concurrent users, variable workloads and latency requirements
2. Security & Governance	Limited access to controlled data/tools	Requires authentication, authorization, data protection, guardrails and policy enforcement
3. Reliability & Quality	Successful demo is enough	Must deliver consistent, accurate and predictable behavior, including handling failures
4. Cost & Efficiency	Cost is usually secondary	Model calls, tokens and infrastructure must be optimized for sustainable cost and performance
5. Operations & Improvement	Developers manually observe and troubleshoot	Requires monitoring, tracing, evaluation and continuous improvement in production

What Does an AI Agent Need Before It Can Run in Production?



Agentic AI on AWS



Enable AI agent integration and continuous improvement

Enforce policies, safety & governance

Provide intelligence for AI applications

Deploy, serve and expose AI models at scale

Maximize Inference efficiency And performance

Provide the compute foundation for AI workloads

- Bedrock AgentCore Harness - Agent orchestration & agent loop
- AgentCore Memory - Short & long-term memory
- AgentCore Gateway - Connect agents to tools
- AgentCore Browser / Code Interpreter - Agent tools

- Bedrock Guardrails - Safety & policy controls
- AgentCore Identity - Authentication & authorization
- AgentCore Policy - Control agent actions
- AgentCore Observability - Monitor agent behavior

- Amazon Nova - AWS foundation models
- Anthropic Claude - Reasoning & agentic models
- Other Bedrock Models - Meta, Mistral and other model providers

- Amazon Bedrock - Serverless model inference
- Amazon SageMaker AI - Model deployment & endpoints
- Amazon EC2 / EKS - Self-managed model serving

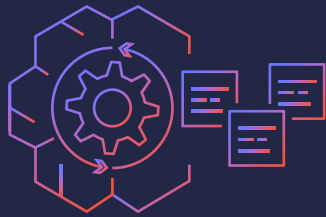
- Prompt Caching - Reduce latency & input-token cost
- Intelligent Prompt Routing - Optimize model selection for quality & cost
- Inference Profiles - Manage & route model inference

- NVIDIA GPU Instances - GPU-accelerated AI
- AWS Inferentia - Purpose-built inference acceleration
- AWS Trainium - AI training & inference
- EC2 UltraClusters - Scale-out accelerated compute

Amazon Bedrock AgentCore

- The platform to build, connect, and optimize agents. Any framework, any model.

SHIP AGENTS FAST



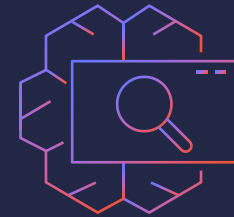
Build and deploy agents
without months of
infrastructure work

CONNECT TO ANYTHING



Give agents secure access
to MCP servers, APIs, and
knowledge bases

OPTIMIZE CONTINUOUSLY



Trace every agent decision,
evaluate its performance,
and safely test improvements
in production

Security at scale

Enterprise readiness

Deterministic control

Everything you need for getting agents to production

Amazon Bedrock AgentCore

AGENT HARNESS

Managed harness
Strands Agents SDK
Any framework, model, or harness

CONTEXT

Memory
Managed Knowledge Base
Web Search

TOOLS

Gateway
Browser
Code interpreter
Payments*

OPTIMIZATION

Observability
Evaluation
Insights*
Recommendations,
A/B testing

ENVIRONMENT

Runtime

PLATFORM CONTROL

AWS Agent Registry*

Identity

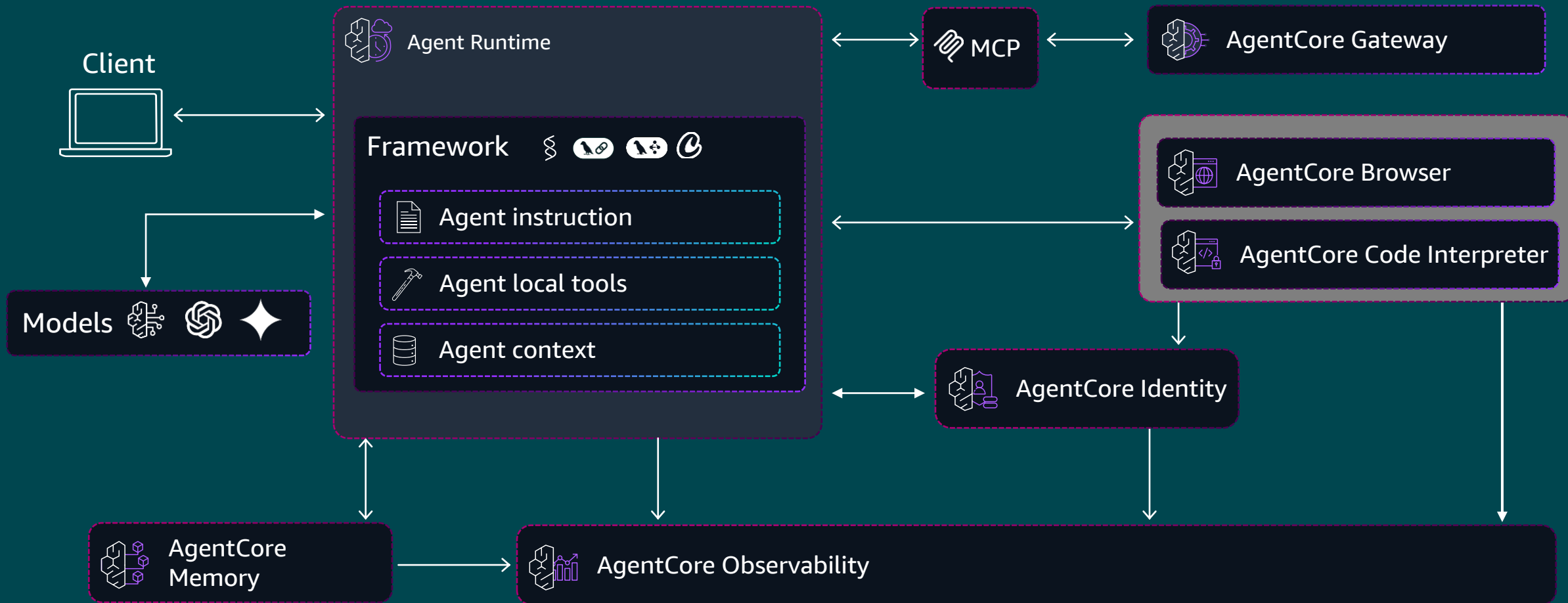
Policy

Guardrails

COMPOSABLE BY DESIGN. PAY ONLY FOR WHAT YOU USE.

*Preview features

Amazon Bedrock AgentCore - Connections



Playlist – Bedrock AgentCore Fundamentals



Amazon Bedrock AgentCore Workshop ...

by BeSA Cloud Academy

Playlist • 8 videos • 2,305 views

Build production-ready AI agents from scratch using Amazon Bedrock AgentCore – the complete hands ...more



Play all



Amazon Bedrock AgentCore Fundamentals | Deploy AI Agents at Scale

BeSA Cloud Academy

1.8k views • 3 months ago



Lab 1: Create Your First AI Agent with Tools – Bedrock AgentCore Workshop

BeSA Cloud Academy

836 views • 3 months ago



Lab 2: Your Agent Forgets Everything? Fix It with AgentCore Memory

BeSA Cloud Academy

262 views • 3 months ago



Lab 3: AgentCore Gateway - One Endpoint, Infinite Agents, Zero Trust

BeSA Cloud Academy

226 views • 3 months ago



Lab 4: From Jupyter Notebook to Production - AgentCore Runtime

BeSA Cloud Academy

125 views • 3 months ago



MATT WOOD



PRASAD RAO



Chief AI & Technology
Officer at AWS



Principal Solutions
Architect at AWS

AGENTIC AI



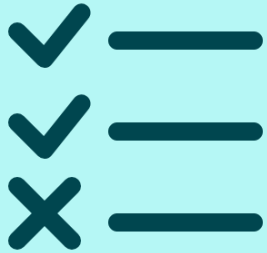


Hands-on Workshop

Walkthrough

Next Steps: Your Turn to Act

1



Complete the
Knowledge
Check

2



Practice the
Hands-on
Workshop

3



Be active and
Support
Your Peers

4



Share Your
Learning on
Social Media

Batch 10 Details

- Live sessions every Saturday (will be recorded)
 - 22-Aug to 26-Sep
- Three Tracks
 - Technical Track –
 - Agentic AI: PoC to Production on AWS
 - Behavioural Track –
 - Leading Customer Discussions as an SA
 - Workshop Track –
 - Hands-on experience through AWS Workshops
- Learning Partners
 - Whizlabs, Packt

